

RESEARCH STUDY - FEBRUARY 2026

AI FOR AMERICANS FIRST

AI Protectionism, Energy and Semiconductors: US/Europe Divergence Trajectories 2024-2030

Integrated Geostrategy and Economic Analysis - GENERAL CONCLUSION

**76.9% of global operational AI
compute = USA**

**1.59x EU/US energy cost (PPP-
adjusted)**

**3.46:1 US/EU CACI ratio (Power
Mode)**

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GENERAL CONCLUSION

From AI Protectionism to the Recomposition of the Global Technological Order

This conclusion synthesizes the results of the study conducted over the 2022-2026 period, validates the central hypothesis of an American AI protectionist regime, outlines the contributions to the literature, identifies limitations and research avenues, and argues the civilizational challenge represented by AI compute as a fourth factor of production. It closes with a summary table of the eleven chapters and an inventory of the primary sources mobilized.

1. Validation of the central hypothesis

This study started from a precise hypothesis: that the Trump 2.0 administration would transform the Biden export controls into a broader protectionist regime, using AI compute as an instrument of economic and geopolitical power—an implicit 'AI for Americans First' decree. The empirical analysis conducted over the 2022-2026 period substantially validates this hypothesis.

On January 15, 2026, the Trump administration simultaneously promulgated a 25 percent tariff (Section 232) on advanced AI semiconductors (Nvidia H200, AMD MI325X) for re-exports to China, and published the BIS final rule governing AI chip exports. The combination of tariffs and export controls constitutes precisely the hybrid mechanism we anticipated: a tax on access to frontier compute that generates revenue for the US

Treasury while slowing down competitors, combined with unlimited domestic access that strengthens the competitive advantage of US Big Tech.[1]

Furthermore, the July 2025 AI Action Plan formalizes a doctrine that goes beyond simple national security controls: exporting the complete AI stack (hardware, models, software, applications, and standards) to countries willing to join the American AI alliance, subject to compliance with US security requirements.[2] AI is no longer treated as just another technology, but as an instrument of power projection analogous to the dollar in the monetary system or oil in the energy system.

2. Synthesis of results

2.1 A measurable and growing American competitive advantage

The CACI (Compute-Adjusted Competitive Index) developed in this study allows for the quantification of structural asymmetry. On the April 2026 public dashboard snapshot, the US/EU(13) operational installed compute raw ratio stands at 17.6:1, translated by a weighted geometric formula $F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$ into a CACI Power Mode ratio of 3.64:1. The concentration of 78.9 percent of global operational AI compute in the United States (49.9 percent including planned capacities), an annual hyperscaler capex of 660-690 billion USD (higher than Sweden's GDP), and a frontier model training cost 5 to 10 times lower than the European cost confirm this dominance. The advantage is self-reinforcing: companies with abundant access to compute capture innovation and data rents that are very difficult to catch up with later (Chapter IV).

An important methodological nuance from Chapter I (Fig 1.8): the Phys/Sov decomposition rigorously calculated from the Epoch AI 'Owner' field reveals that asymmetry is differentiated by jurisdiction. The United States and China are fully sovereign over their installed compute (CACI Phys = CACI Sov). The EU is also very largely sovereign over its installed F (78.6 percent EU-owned). The extreme case is that of the UAE: 99.6 percent of Emirati F_total is held by US-side operators (Stargate UAE, Microsoft, OpenAI), causing the sovereign CACI to drop from 55.7 (Physical) to 6.0. This Phys/Sov dissociation, formalized in Chapter V section 5.9.2 under the hypothesis of the 2028 Cloud Sovereignty Mandates activation, radically transforms the reading of asymmetry: European vulnerability lies not on installed compute but on the operational layer of cloud workloads (mostly hosted on AWS/Azure/GCP).

2.2 A three-tier protectionist architecture

The analysis reveals that American AI protectionism operates at three distinct but cumulative levels.

Tier 1: Export controls (inherited from Biden, maintained and transformed by Trump). The tier system (Tier 1/2/3) segments the world based on geopolitical alignment: free access for close allies (20 countries), quantitative caps for the rest of the world, prohibition for adversaries (China, Russia). Even after the formal repeal of the AI Diffusion Rule in May 2025, regulatory uncertainty weighs on the investment decisions of Tier 2 countries (Chapter III).

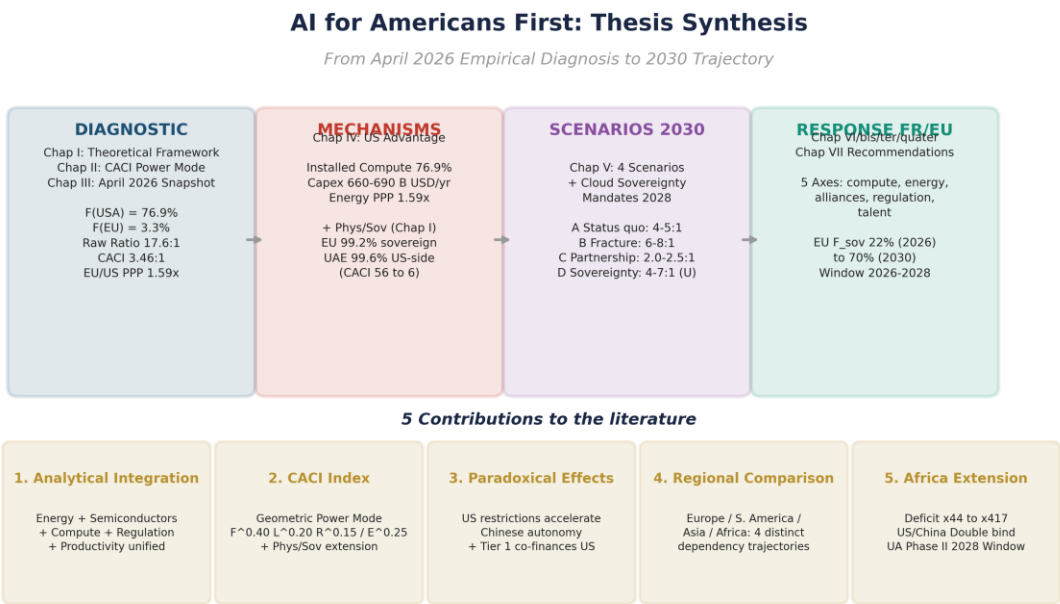
Tier 2: Customs tariffs (Trump innovation). The 25 percent tariff on advanced AI semiconductors (Section 232, January 2026) constitutes a rupture: export controls targeted national security, tariffs explicitly target revenue and competitive advantage. The combination of tariffs and domestic exemptions creates a direct cost differential between American and non-American companies (Chapter V).

Tier 3: Capitalistic gravity. The concentration of capex (660-690 billion USD across five companies in 2026), combined with energy access (the United States accepts increased use of fossil fuels, 53.7 GW of installed DC capacity), creates a gravitational effect: investments from Japan (550 billion), the UAE, and SoftBank/Stargate converge on American soil, strengthening the compute hub without additional regulatory intervention (Chapter VI ter).

A potential fourth tier is emerging on the 2028 horizon: the Cloud Sovereignty Mandates analyzed in Chapter V section 5.9. By extending the requirements of the Framework for AI Diffusion to the cloud layer, they would transform US hyperscalers operating offshore into conditional intermediaries of global compute. The European vulnerability window (operational CADA by 2027-2028 at best, US Mandates activatable in 2028) is maximal between 2028 and 2030.

2.3 Differentiated consequences by region

Table 24 below summarizes the structural positions and specific risks of each region studied, integrating the extension to Africa developed in Chapter VI quater.



Source: Doctoral thesis Fabrice Pizzi (Université Paris-Sorbonne, M2 IE). Dashboard: <https://moogly.github.io/America-First-IA/dashboard/>

3. Contributions of this study

This research makes five contributions to the economic and geostrategic literature.

First, the analytical integration of trajectories usually treated separately—energy, semiconductors, compute, regulation, productivity—into a unified framework. Most academic works treat these dimensions separately; our analysis shows that they form a system of interdependencies where each constraint amplifies the others (energy constrains compute, compute constrains productivity, productivity determines competitiveness).

Second, the proposal of the CACI (Compute-Adjusted Competitive Index), which provides a measurement framework for comparing AI competitiveness between regions by integrating available FLOPs, human capital, regulation, and energy cost according to a weighted geometric formula. While this index remains to be empirically refined, it constitutes a first attempt to synthesize the concept of compute-adjusted competitiveness identified as missing in the literature (Chapter II). The Phys/Sov extension introduced in Chapter I and formalized in Chapter V ($F = F_{\text{phys}} \times F_{\text{sov}}$) adds a jurisdictional dimension that distinguishes physically present compute from legally controllable compute—a distinction operational as of the April 2026 snapshot (UAE 99.6 percent US-side case) and systemic under the 2028 Cloud Sovereignty Mandates regime.

Third, the demonstration that American AI protectionism produces systemic paradoxical effects. Restrictions intended to maintain the US advantage accelerate the construction of an alternative Chinese ecosystem (DeepSeek, Huawei Ascend, real capacity 246-300 EFLOP/s versus 0.5 percent apparent in consolidated Epoch AI data), push Tier 2 countries toward China (ByteDance in Brazil, ASEAN, Africa), and incentivize Tier 1 allies to co-finance US supremacy rather than build true autonomy (Japan: 550 billion to the United States). AI protectionism does not produce a unipolar world but a world fragmented into technological blocs.

Fourth, the unprecedented comparative analysis of regional responses to AI protectionism (Europe, South America, Asia, Africa), showing that geopolitical position, energy endowment, and proximity to value chains determine fundamentally different dependency trajectories, irreducible to a single model of catch-up or falling behind.

Fifth, the extension to Africa (Chapter VI quater) documents the most extreme compute asymmetry in the world (deficit x44 to x417 depending on indicators) and shows how American protectionism creates a specific double bind for this continent: restriction of access to US frontier compute on one side, exposure to risks of surveillance and secondary sanctions from using the Chinese alternative on the other.

4. Limitations and research avenues

This study has several limitations that should be explicitly stated.

Regulatory uncertainty. The export controls environment is evolving rapidly. Biden's AI Diffusion Rule was repealed in May 2025; the Trump final rule of January 2026 could itself be modified (Commerce is to provide an update to the President by July 2026). The scenarios proposed in Chapter V reflect this uncertainty, but the space of possibilities is broader than the four formalized scenarios.

Fragmentary data. AI compute data by region remains incomplete despite the rigorous April 2026 public dashboard snapshot. Epoch AI estimates significantly under-represent real Chinese capacity (China 0.5 percent apparent vs 246-300 EFLOP/s claimed) due to the anonymization of Chinese clusters and the opacity

of Huawei/Cambricon/Biren providers. The CACI is an exploratory index, calibrated on the April 2026 snapshot but not yet validated over long time series.

Time horizon. The analysis covers 2026-2030, but technological disruptions (quantum computing, sub-2 nm nodes, neuromorphic architectures) could redistribute the cards after 2030. Nvidia's current GPU advantage could be challenged by specialized ASICs (Google TPU, Amazon Trainium, Huawei Ascend) or radically different architectures (European DARE/RISC-V, 2030-2032 horizon).

Sensitivity to CACI weightings. The weightings of the geometric formula ($F^{0.40} \times L^{0.20} \times R^{0.15} / E^{0.25}$) were chosen in Chapter II based on the literature but do not stem from an econometric calibration. A systematic sensitivity analysis on these weightings could reveal alternative trajectories not explored.

Future research avenues. Four extensions are required. First, the empirical calibration of the CACI on survey data (sectoral productivity by compute access) would allow for the validation or adjustment of current weightings. Second, the sectoral deepening of African coverage (Chapter VI quater)—notably the country-by-country analysis of the 16 identified national AI strategies and the implementation of the AU Continental Strategy Phase II 2028. Third, the dynamic modeling of the energy-compute-productivity interaction via computable general equilibrium (CGE) models integrating compute constraints as a production factor. Fourth, the longitudinal observation of the 2028 Cloud Sovereignty Mandates regime (if it actually activates) and its effects on the F_{sov} trajectory of different jurisdictions.

5. The civilizational challenge

Beyond economic metrics and geopolitical scenarios, this study reveals a more fundamental challenge. AI compute is becoming the fourth factor of production (after capital, labor, and land/energy), structuring access to productivity gains, innovation, and ultimately prosperity. Like oil in the 20th century, the control of compute in the 21st century will determine which nations and which companies capture the rents of innovation.

The United States has understood this. The July 2025 AI Action Plan explicitly treats the AI stack as an instrument of geopolitical alliance, comparable to the Marshall Plan or the Bretton Woods system: access to American compute is conditioned on strategic alignment, creating a system of hierarchical dependencies. Carnegie notes that the rule aimed to use AI exports as leverage over geopolitical pivot states, by establishing incentives for other governments to adopt American technological standards and protections in exchange for US chips.[3]

Faced with this new system, France and Europe have a strategic choice that fundamentally boils down to three options. The first is subordinate integration: accepting the status of technological junior partner in the American bloc, as Japan chose by investing 550 billion USD on US soil. This option minimizes the risk of access disruption but maximizes dependency. The second is sovereigntist confrontation: building an entirely autonomous AI ecosystem, as China is forced to do. This option is unrealistic by 2030 for Europe, which lacks both a sufficient semiconductor industrial base and interior market capacity.

The third option—the one recommended by this study in Chapter VII—is targeted strategic autonomy. it consists of building sovereignty on the segments where Europe possesses a comparative advantage (French nuclear energy at 1.35x US PPP cost, ASML lithography equipment, Mistral open AI models, AI Act regulatory framework) while maintaining interoperability with the American ecosystem. The goal is not autarky but the capacity of choice: to have credible alternatives (SOV-3 sovereign cloud, local compute under EU jurisdiction, open models) to never be captive to a provider whose geopolitical interests might diverge from ours. The Phys/Sov distinction established in Chapter I is operational here: it is about increasing F_{sov} on the cloud workload layer, not just F_{phys} on the installed infrastructure.

Time is of the essence. The tipping point identified in this study is in 2028: convergence of EU compute and energy saturation (Chapter V section 5.7.3), potential activation of Cloud Sovereignty Mandates (Chapter V section 5.9), and the probable end of the vulnerability window before positions crystallize. After this date, positions stiffen around the 17.6:1 raw / 3.64:1 CACI Power Mode baseline and dependencies become structural. The 2026-2028 strategic action window is narrow. The 109 billion EUR in AI investments announced for France, the 200 billion EUR InvestAI program, the rise of Mistral Compute, and dedicated EDF nuclear sites are the elements of a response. But between announcement and execution, there is the distance that separates strategy from reality. India promises 200 billion USD but only has 1.4 GW installed. Europe cannot afford a comparable gap between ambition and realization.

Ultimately, 'AI for Americans First' is not just a trade policy scenario. It is the signal of a recomposition of the global technological order comparable to the major restructurings of the 20th century—Bretton Woods, the oil shock, the end of the Cold War. Each of these ruptures created winners and losers for decades. The question for France and Europe is no longer whether this recomposition will take place—it is underway—but to determine whether we will be its architects or its subjects.

Fabrice Pizzi, Paris, February 2026.

Table 24. Summary of regional consequences of American AI protectionism.

Source: Author's construction, calibrated on the April 2026 snapshot (US 78.9% operational AI compute, EU raw ratio 17.6:1, CACI Power Mode 3.64:1).

Region	Structural Position	Main Impact	Specific Risk
Europe / France	Tier 1, dependent on US GPU + cloud (72-80% workloads)	Compute gap 17.6:1 raw / 3.64:1 CACI; training costs x5-10	Geopolitical vendor lock-in; marginalization if US-Asia bloc closes; F _{sov} vulnerability on cloud workloads
South America / Brazil	Tier 2, US-China competition ground	Technological bifurcation; amplified brain drain	Triple fracture (North-South, East-West, intra-regional)
Japan / Korea / Taiwan	Tier 1, critical value chain links	Co-financing US supremacy (550B USD Japan); production transfer	Asymmetric partnership; Taiwan advantage erosion; Japanese investment in US instead of EU
India	Tier 2, Global South pivot	GPU cap tension vs compute hub ambition	Applicative sovereignty without hardware sovereignty
China	Tier 3, forced	Parallel AI ecosystem	Permanent technological

	autonomization	(Huawei/DeepSeek); real capacity 246-300 EFLOP/s; 2-3 gen GPU lag	bifurcation; exporting to Tier 2/3 (Brazil, ASEAN, Africa)
Africa	Tier 2/3, compute deficit x44-x417	Extreme asymmetry; US/China double bind	Huawei/DeepSeek dependency; surveillance; structural confinement; UAE 99.6% US-side case

Table 25. Summary of chapters, volume, and critical apparatus of the study.

Source: Author's construction. Volume (in indicative pages) includes figures and tables but excludes econometric appendices.

Chapter	Title	Indicative Pages	Notes
I	Theoretical Framework: Technological Protectionism and AI	~12	22
II	Methodology: Scenario Matrix and CACI Index	~8	10
III	Empirical Diagnosis 2020-2026: Energy, Semiconductors, Compute	~11	20
IV	Mechanisms of the US Competitive Advantage	~9	19
V	Prospective Scenarios 2026-2030 and Cloud Sovereignty Mandates	~14	29
VI	Consequences for France and Europe	~10	14
VI bis	Consequences for South America and Brazil	~11	19
VI ter	Consequences for Asia	~12	16
VI quater	Consequences for Africa	~13	26
VII	Strategic Recommendations for France and Europe	~11	18
Conclusion	From AI Protectionism to the Recomposition of the Global Technological Order	~9	3
TOTAL	11 chapters	~120	196

Primary sources mobilized: IEA, McKinsey, Bruegel, Brookings, Carnegie Endowment, European Commission, White House/BIS, European Parliament, CSIS, S&P Global, Epoch AI, Centre for Future Generations (CFG), Euronews, ECLAC/CENIA (ILIA 2025), World Bank, Futurum, Introl, World Nuclear News, Arizton, Pillsbury Law, ITIF, Foreign Policy, Hudson Institute, Mordor Intelligence, McKinsey Global Institute, IMF. Additional data: Bloomberg, DCD, Morgan Lewis, Tom's Hardware, Serrari Group, Data Center Knowledge, WEF, Africa Defense Forum, Atlantic Council DFRLab, Carnegie Endowment, New Lines Institute, RTE, EDF, ANSSI, USTDA. Primary data: Epoch AI public dashboard snapshot April 2026 (<https://mo0ogly.github.io/America-First-IA/dashboard/>).

Notes

1. Pillsbury Law (January 2026), 'Trump Admin Targets Advanced AI Semiconductors'. Section 232: 25 percent tariff on Nvidia H200, AMD MI325X for China re-export. US domestic exemptions. Simultaneous BIS final rule. DC market update scheduled July 2026.

2. White House / CM Trade Law (July 2025), 'America's AI Action Plan'. Pillar III: export the full AI technology stack to allies. Four principles: export to allies, enforcement strengthening, global alignment, measures protection.

3. Carnegie Endowment for International Peace (May 2025), 'The Trump Administration May Be About to Repeal the AI Diffusion Rule'. Analysis of the control/promotion/leverage trilemma. Recommendation: broaden Tier 1 group, increase India allocations, strengthen localization requirements.

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Public Dashboard : <https://moOogly.github.io/America-First-IA/dashboard/>

Repository : <https://github.com/moOogly/America-First-IA>

AI for Americans First - Fabrice Pizzi - General Conclusion